ECE 402 Communications Engineering
Department of Electrical and Computer Engineering
NC State University
Fall 2026

Instructor: Prof. David Huaiyu Dai, EB2 2078, hdai@ncsu.edu, Office Hours: Tu-Th 2:45–3:45pm and by appointment. TA contact info and office hours will be posted later on the website.

Text: J. G. Proakis and M. Salehi, *Fundamentals of Communication Systems,* 2nd ed. Englewood Cliffs, NJ: Prentice Hall, 2014. The text will be delivered electronically on Moodle. Access will be free until the census date (8/28). However, you must purchase it by the census date; otherwise you will lose access.

Description: An overview of digital communications for wireline and wireless channels which focuses on reliable data transmission in the presence of bandwidth constraints and noise. The emphasis is on the unifying principles common to all communications systems. Examples include cellular telephony, local-area networks, high-speed modems and satellite communications.

General: Prereqs: ECE 301, ST 371. Lectures are Tu-Th 1:30–2:45pm in EBII 1228. Classes will also be captured automatically and made available on the website for those who cannot attend. However, taking the class remotely is strongly discouraged. Lecture notes, homework assignments, readings, practice exams and homeworks, and message boards are available on the website, `http://wolfware.ncsu.edu/`. There will be a total of 8 homework assignments and 4 exams. Solutions will be posted on the website. The lowest exam and homework scores will be dropped. Late homeworks will not be graded. Requests for regrading must be made within one week after an assignment is returned.

Projects: Each homework will have one "project" question that involves the use of MATLAB. These questions explore the impact of digitization, noise and bandwidth constraints on communication system performance. Your understanding of the projects will be tested both on the homeworks and the exams.

MATLAB: All NC State students can install MATLAB at no cost on personal computers (see link at top of course website). Be sure to install version R2024b (or later) together with the Communications, DSP System and Signal Processing Toolboxes, which are needed for some projects. Instructions for installing these toolboxes are on the course website.

Exams: There are 3 quizzes and one cumulative final exam. During exams, the only reference permitted is a class formula sheet; otherwise, exams are closed book and class materials. Conventional engineering calculators may be used. The lowest exam grade is dropped. Excused absences on exams require documentation; make-up exams will be scheduled only for students with more than one excused absence.

Grades: Homework 20% (best seven of eight), exams 80% (best three of four). Current grades will be available in the online gradebook on the course website. The nominal grade cutoffs are: A+ 97%, A 93%, A- 90%, B+ 87%, B 83%, B- 80%, C+ 77%, C 73%, C- 70%, D+ 67%, D 63%, D- 60%. Rarely, the final grade may be curved.

Academic Integrity: All students are expected to follow the NCSU Code of Student Conduct. For more information, visit `https://policies.ncsu.edu/policy/pol-11-35-01`. You may discuss homework assignments with other students, but you may not copy or transcribe their work. Similarly, you may not copy or transcribe solutions found on the internet. Artificial Intelligence can be a useful learning tool, but does not replace the engineering thought process that this class emphasizes, and you may not copy or transcribe solutions provided by AI. It is an academic integrity violation to submit solutions from other students, the internet or AI tools as your own, without proper authorization and attribution.

Disabilities: Reasonable accommodations will be made for students with verifiable disabilities. To qualify, students must register with Disability Resources at `http://dro.dasa.ncsu.edu/`. Exams taken at DRO must significantly overlap (> 50%) the scheduled class exam. You are responsible for checking the overlap when you schedule the test. We strongly recommend scheduling all exams to be taken at DRO during the first week of classes.

References: The following books have been placed on reserve at the library:

1. S. Haykin and M. Moher, *Communication Systems,* 5nd Ed. New York: Wiley, 2009.
2. J. G. Proakis and M. Salehi, *Fundamentals of Communication Systems,* 2nd ed. Englewood Cliffs, NJ: Prentice Hall, 2014.

ECE 402: Fall 2026 Lectures and Assignments

(tentative)

Date	H	Topic	Reading
8/18		Overview	1.1–1.4
8/20		Fourier Series	2.1–2.2
Project			
8/25		Fourier Transforms	2.3
8/27		Linear Systems	2.3
Project			
9/1		Energy/Power Spectral Densities	2.5
9/3	H1	Sampling and Quantization	7.1–7.2.1
Project		P1–1.1: Working with MATLAB	
9/8		Pulse Code Modulation	7.3–7.4.1
9/10	H2	Probability Review	5.1
Project		P1–1.2: Energy & Power Spectra	
9/15		Random Processes	5.2–5.3
9/17		White Gaussian Noise	
Project			
9/22	H3	Quiz I Problem Session	
9/24		**Quiz I**	
Project		P1–1.3: Sampling & Quantization	
9/29		**Wellness Day: No Class**	
10/1		Digital Baseband Modulation	8.1–8.2
Project			
10/6		Optimal Receivers	8.3
10/8	H4	Probability of Error	8.4
Project		P2–1.1: Digital Data Transmission	

All readings are in Proakis and Salehi

Date	H	Topic	Reading
10/13		Bandpass Modulation	8.5–8.7
10/15		Bandwidth and Power Efficiency	9.7
Project			
10/20		**Fall Break: No Class**	
10/22		Channel Capacity	12.6
Project			
10/27	H5	Communication Tradeoffs	
10/29		Space/Satellite Communications	14.5
Project		P2–1.2: Communications Tradeoffs	
11/3	H6	Quiz II Problem Session	
11/5		**Quiz II**	
Project		P2–1.3: Satellite Link Budgets	
11/10		Intersymbol Interference	10.1
11/12		Nyquist's Criterion	10.3
Project			
11/17	H7	Raised-Cosine Spectra	10.3
11/19		Equalization	10.5.2
Project		P3–1.1: Digital Data Transmission	
11/24		Multicarrier Modulation/DSL	11.1,2,3,6
11/26		**Thanksgiving: No Class**	
Project			
12/1	H8	Quiz III Problem Session	
Project		P3–1.2: Visualizing ISI	
12/3		**Quiz III & Final**	
		Thursday 12–2:30pm	